

1 Comparing Language Input in Homes of Young Blind and Sighted Children: Insights from
2 Daylong Recordings

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Abstract

We compared everyday language input to young congenitally-blind children with no additional disabilities (N=15, 6–30mo., M:16mo.) and demographically-matched sighted peers (N=15, 6–31mo., M:16mo.). By studying whether the language input of blind children differs from their sighted peers, we aimed to determine whether, in principle, the language acquisition patterns observed in blind and sighted children could be explained by aspects of the speech they hear. Children wore LENA recorders to capture the auditory language environment in their homes. Speech in these recordings was then analyzed with a mix of automated and manually-transcribed measures across various subsets and dimensions of language input. These included measures of quantity (adult words), interaction (conversational turns and child-directed speech), linguistic properties (lexical diversity and mean length of utterance), and conceptual features (talk centered around the here-and-now; talk focused on visual referents that would be inaccessible to the blind but not sighted children). Overall, we found broad similarity across groups in speech quantitative, interactive, and linguistic properties. The only exception was that blind children’s language environments contained slightly but significantly more talk about past/future/hypothetical events than sighted children’s input; both groups received equivalent quantities of “visual” speech input. The findings challenge the notion that blind children’s language input diverges substantially from sighted children’s; while the input is highly variable across children, it is not systematically so across groups, across nearly all measures. The findings suggest instead that blind children and sighted children alike receive input that readily supports their language development, with open questions remaining regarding how this input may be differentially leveraged by language learners in early childhood.

Introduction

The early language skills of blind children are highly variable. Some children demonstrate age-appropriate vocabulary and grammar from the earliest stages of language learning, while others experience substantial language delays (Bigelow, 1987; E. E. Campbell, Casillas, & Bergelson, 2024; Landau & Gleitman, 1985). By adulthood, however, blind individuals are fluent language-users, even demonstrating faster lexical processing skills than sighted adults (Loiotile, Lane, Omaki, & Bedny, 2020; Röder, Demuth, Streb, & Rösler, 2003; Röder, Rösler, & Neville, 2000; though c.f., Sak-Wernicka, 2017 for discussion of possible pragmatic differences). The causes of early variability and the potential ability (or need) to “catch up” remain poorly understood: what could make the language learning problem different or initially more difficult for the blind child? Here, we compare the language environments of blind children to that of their sighted peers. In doing so, we begin to untangle the role that perceptual input plays in shaping children’s language environment, and better understand the interlocking factors that may contribute to variability in blind children’s early language abilities.

Why would input matter?

Among both typically-developing children and children with developmental differences, language input has been found to predict variability in language outcomes (Anderson, Graham, Prime, Jenkins, & Madigan, 2021, 2021; Gilkerson et al., 2018; Huttenlocher, Haight, Bryk, Seltzer, & Lyons, 1991; Huttenlocher, Waterfall, Vasilyeva, Vevea, & Hedges, 2010; Rowe, 2008, 2012). At a coarse level, children who are exposed to more speech (or sign, Watkins, Pittman, & Walden, 1998) tend to have stronger language outcomes and produce more speech themselves (Anderson et al., 2021; Bergelson et al., 2023; Gilkerson et al., 2018; Huttenlocher et al., 1991; Rowe, 2008).

Previous research suggests that the structure and content of the language input (often

referred to as input “quality”¹ is even more influential than the amount of speech alone (Hirsh-Pasek et al., 2015; Rowe, 2012). Rowe and Snow (2020) categorized the makeup of the input along three dimensions: interactive features (e.g., parent responsiveness, speech directed *to* child vs. overheard, conversational turn-taking), linguistic features (e.g., lexical diversity, grammatical complexity), and conceptual features (i.e., the extent to which input focuses on the *here-and-now*).

In examining interactive features, previous studies have indicated that back-and-forth communicative exchanges (also known as conversational turns) between caregivers and children are predictive of better language outcomes across infancy (Donnellan, Bannard, McGillion, Slocombe, & Matthews, 2020; Goldstein & Schwade, 2008) and toddlerhood (Hirsh-Pasek et al., 2015; Romeo et al., 2018). Another way to quantify caregiver and infant interaction is by looking at how much speech is directed *to* the child (e.g. as opposed to an overheard conversation between adults). The amount of child-directed speech in children’s input (at least in Western contexts, Casillas, Brown, & Levinson, 2020) has been linked to children’s vocabulary size and lexical processing (Rowe, 2008; Shneidman, Arroyo, Levine, & Goldin-Meadow, 2013; Weisleder & Fernald, 2013).

Under the linguistic umbrella, we can measure the *kinds* of words used (often measured as lexical diversity, type-token ratio), and the ways they are *combined* (syntactic complexity, often measured by mean length of utterance). Both parameters have been found to correlate with children’s language growth: sighted toddlers who are exposed to a greater diversity of words in their language input are reported to have larger vocabulary scores (Anderson et al., 2021; Hsu, Hadley, & Rispoli, 2017; Huttenlocher et al., 2010; Rowe, 2012; Weizman & Snow, 2001). Likewise, the diversity and complexity of syntactic constructions in parental language input has been associated with both children’s vocabulary growth and structural diversity in

¹ We avoid the term “quality” here as it carries potential biases regarding linguistic norms (MacLeod & Demers, 2023).

their own productions (De Villiers, 1985; Hadley et al., 2017; Hoff, 2003; Huttenlocher, Vasilyeva, Cymerman, & Levine, 2002; Huttenlocher et al., 2010; Naigles & Hoff-Ginsberg, 1998).

Finally, the conceptual dimension of language input aims to capture the extent to which the language signal maps onto present objects and ongoing events in children’s environments (Rowe & Snow, 2020). As children develop, their ability to represent abstract referents improves (Bergelson & Swingley, 2013; Kramer, Hill, & Cohen, 1975; Luchkina, Xu, Sobel, & Morgan, 2020). Decontextualized language input— that is, talking about past, future, or hypothetical events, or people and items that are not currently present in the environment— may be one contributing factor (Rowe, 2013). Greater prevalence of decontextualized language in input to toddlers has been found to predict aspects of children’s own language in kindergarten and beyond (Demir, Rowe, Heller, Goldin-Meadow, & Levine, 2015; Rowe, 2012; Uccelli, Demir-Lira, Rowe, Levine, & Goldin-Meadow, 2019).

From this (necessarily abridged) review, it appears that many factors in the language input alone link to how sighted children learn about the world and language, but that children also learn from sensory, conceptual, and social knowledge. Many cues for word learning are visual: for example, empirical work finds that sighted children can leverage visual information like parental gaze, shared visual attention (Tomasello & Farrar, 1986), pointing (Lucca & Wilbourn, 2018), and the presence of salient objects in the visual field (Yu & Smith, 2012). Because these visual cues are inaccessible to blind children, language input may take on a larger role in the discovery of word meaning (E. E. Campbell & Bergelson, 2022). Syntactic structure in particular provides critical cues to word meaning, such as the relationship between two entities that aren’t within reach, or are intrinsically unobservable or ambiguous (L. Gleitman, 1990). But in order to evaluate whether language input plays a larger role for blind versus sighted children’s learning, it is worth first establishing whether blind and sighted children’s language input differs. That is, children with different sensory

access could differentially make use of the same kind of language input, or they could apply the same learning mechanisms to input with different properties—a debate carried over from work with typically-sighted children (Newport, Gleitman, & Gleitman, 1977). Either way, characterizing the input across potentially relevant dimensions is a helpful first step.

Why would the input differ between blind and sighted children?

Speakers regularly tailor their speech to communicate efficiently with the listener (Grice, 1975). Across many contexts, research finds that parents are sensitive to their child’s developmental level and tune language input accordingly (Newport et al., 1977; Snow, 1972; Vygotsky & Cole, 1978). One example is child-directed speech, wherein parents speak to young children with exaggerated prosody and slower speech (Bernstein Ratner, 1984; Fernald, 1989; Moser et al., 2022; Newport et al., 1977), which are in some cases helpful to the young language learner (Thiessen, Hill, & Saffran, 2005). For instance, parents tend to repeat words more often when interacting with infants than with older children or adults (Fernald & Morikawa, 1993; Snow, 1972). Communicative tailoring is also common in language input to children with disabilities, who have been found to receive simplified, more directive language input, and less interactive input compared to typically-developing children (Dirks, Stevens, Kok, Frijns, & Rieffe, 2020; Yoshinaga-Itano, Sedey, Mason, Wiggin, & Chung, 2020). In other contexts, language input to children with disabilities has been shown to be more multimodal, such that parents more frequently combine communicative cues (e.g., speech and touch, Abu-Zhaya, Kondaurova, Houston, & Seidl, 2019) when interacting with deaf children, compared to their typically-hearing peers.

In addition to tailoring communication to children’s developmental level, speakers also adjust their conversation in accordance to their conversational partner’s sensory access (Gergle, Kraut, & Fussell, 2004 for adults; and Grigoroglou, Edu, & Papafragou, 2016 for adults and 4–6-year-old children). For example, in a noisy environment, adults will often adapt the acoustic-phonetic features of their speech to make it easier for their interlocutor to

understand them (Hazan & Baker, 2011), demonstrating sensitivity to even temporary sensory conditions. When describing scenes, adult speakers tend to provide the information their listeners lack but seem to avoid redundant visual description (Grice, 1975; Ostarek, Paridon, & Montero-Melis, 2019). During in-lab tasks with sighted participants, participants in several studies verbally provide visually-absent cues when an object is occluded to their partner (Hawkins, Gweon, & Goodman, 2021; Jara-Ettinger & Rubio-Fernandez, 2021; Rubio-Fernandez, 2019). These results suggest that adults (Gergle et al., 2004; Hazan & Baker, 2011), children (e.g., Grigoroglou et al., 2016), and even infants (Chiesa, Galati, & Schmidt, 2015; Ganea et al., 2018; Senju et al., 2013) can flexibly adapt communication to the visual and auditory abilities of their partner.

Taking these results into consideration, and given the strong verbal abilities of blind adults (Loiotile et al., 2020; Röder et al., 2003; Röder et al., 2000), we might expect parents of blind children to verbally describe visual information in the child’s environment or otherwise structure interactions to align with their child’s strengths and abilities. But prior research doesn’t yield a clear answer on whether sighted parents modify language input to blind children. Several studies suggest differences in the conceptual features: caregivers of blind children restrict conversation to things that the blind child is currently engaged with, rather than attempt to redirect their attention to other stimuli (Andersen, Dunlea, & Kekelis, 1993; J. Campbell, 2003; Kekelis & Andersen, 1984; though c.f., Moore & McConachie, 1994). Studies of naturalistic input to blind children report that parents use *fewer* declaratives and *more* imperatives than parents of sighted children, suggesting that blind children might be receiving less description than sighted children (Kekelis & Andersen, 1984; Landau & Gleitman, 1985; though c.f., Lukin, Campbell, Richter, & Bergelson, 2023; Pérez-Pereira & Conti-Ramsden, 2001). Other studies report that parents adapt their interactions to their children’s visual abilities, albeit in specific contexts. Tadić, Pring, and Dale (2013) find that in a structured book-reading task, parents of blind children provide more descriptive utterances than parents of sighted children. Further, parents of blind

children have been found to provide more tactile cues to initiate interactions or establish joint attention (Preisler, 1991; Urwin, 1983, 1984), which may serve the same social role as shared gaze in sighted children and take advantage of children’s access to other senses (e.g., touch). These mixed results suggest that parents of blind children might alter language input in some domains but not others. The apparent conflict in results may be exacerbated by the difficulty of recruiting specialized populations to participate in research: the small (in most cases, single-digit) sample sizes of prior work limits our ability to generalize about any differences in the input to blind vs. sighted infants.

The Present Study

Children can and do learn language in a variety of input scenarios (L. R. Gleitman & Newport, 1995), but if language input differs systematically between blind infants and toddlers, capturing this variation may reveal a more nuanced picture of how infants use the input to learn language. In the present study, we examine daylong recordings of the naturalistic language environments of blind and sighted children in order to characterize the input to each group. Using both automated measures and manual transcription of these recordings, we analyze several characteristics that have been previously suggested to be information-rich learning cues, including input quantity (adult word count), interaction (conversational turn count, proportion of child-directed speech), conceptual features (temporal displacement, sensory modality), and linguistic complexity (type-token ratio and mean length of utterance). Though the present study is largely exploratory, we took the directionality of previously reported results as our (admittedly limited) starting point. Thus, based on prior research, we made the tentative predictions that blind vs. sighted children would have input featuring less interactivity [fewer conversational turns and less child-directed speech; Charity Rowland (1984); Grumi et al. (2021)], less linguistic complexity (Bulk, Smith, Nimmon, & Jarus, 2020; lower type-token ratio and shorter utterances, Chernyak, n.d.; Dirks et al., 2020; FamilyConnect, n.d.; Lorang, Venker, &

Sterling, 2020), and conceptual content focused more on the child’s locus of attention (more here-and-now speech and fewer visual words, Andersen et al., 1993; J. Campbell, 2003; Kekelis & Andersen, 1984); we have no *a priori* hypotheses regarding adult word count.

Methods

Table 1

Demographic characteristics of the blind and sighted samples. For continuous variables, range and mean are provided. For categorical variables, percentages by level are provided.

Variable	Blind (N=15)	Sighted (N=15)
Age (months)	6–30, 15.8 (8.2)	6–32, 16.1 (8.1)
Sex	Female: 44%	Female: 44%
	Male: 56%	Male: 56%
Number of Older Siblings	0–2, 0.5 (0.8)	0–3, 1.1 (1)
Race	American Indian or Alaska Native: 6%	American Indian or Alaska Native: 0%
	Black or African American: 6%	Black or African American: 6%
	Mixed: 19%	Mixed: 6%
	White: 69%	White: 56%
	unknown: 0%	unknown: 31%
Maternal Education	Some college: 19%	Some college: 6%
	Associate’s degree: 6%	Associate’s degree: 12%
	Bachelor’s degree: 31%	Bachelor’s degree: 56%
	Graduate degree: 44%	Graduate degree: 6%
Ethnicity	Hispanic or Latino: 19%	Hispanic or Latino: 0%
	Not Hispanic or Latino: 81%	Not Hispanic or Latino: 62%
	unknown: 0%	unknown: 38%

Participants

This study included 15 congenitally-blind infants and their families². To be eligible, participants had to be 6–30 months old, have severe to profound visual impairment (i.e. at most light perception), no additional disabilities (developmental delays, intellectual

² One family contributed two recordings for the same blind child. In the present study, we used only the first recording from that participant.

disabilities, or hearing loss), and be exposed to $\geq 75\%$ English at home. Blind participants were recruited through ophthalmologist referral, preschools, early intervention programs, social media, and word of mouth. Blindness in our sample was caused by a range of conditions, including cataracts (n=3), Leber’s Congenital Amaurosis (n=1), Microphthalmia (n=2), Ocular albinism (n=2), Optic Nerve Hypoplasia (n=2), Retinal Detachments (n=1), and Retinopathy of Prematurity (n=1). Etiology was unknown in 2 participants, and 2 participants had multiple contributing conditions. Caregivers were also asked to complete a demographics survey and the MacArthur-Bates Communicative Development Inventory (CDI, Fenson et al., 1994) within one week of the home language recording.

To control for the wide age range of the study, each blind participant was matched to a sighted participant, based on age (± 6 weeks), sex, maternal education ($\pm$ one education level), and number of siblings (± 1 sibling). Sighted matches were drawn from multiple existing corpora: two children from VanDam et al. (2015) and VanDam et al. (2016); five children from Bergelson (2015) and Bergelson, Amatuni, Dailey, Koorathota, and Tor (2019); one child from Ramírez-Esparza, García-Sierra, and Kuhl (2014); two children from Warlaumont, Pretzer, Mendoza, and Walle (2016); two from Wang, Cooke, Reed, Dilley, and Houston (2022); and two from Caroline Rowland et al. (2018)³. There was no recording available that matched two blind participant’s demographic characteristics; we therefore collected recordings from two sighted children *de novo*. See Table 1 for sample demographic characteristics.

Recording Procedure

For the recording portion of the study, caregivers of participating infants received a LENA wearable audio recorder and vest (Ganek & Eriks-Brophy, 2016; Gilkerson &

³ These two sighted children are from the UK, the rest from North America. While recognizing this potential limitation, we have no *a priori* reason to predict that North American and UK English learners should differ meaningfully in our language measures, especially given our broader demographic matching procedure.

Richards, 2008). They were instructed to place the recorder in the vest on the day of their scheduled recording and put the vest on their child from the time they woke up until the recorder automatically shut off after 16 hours (setting the vest nearby during baths, naps, and car rides). Actual recording length ranged from 8 hours 17 minutes to 15 hours 59 minutes (Mean: 15 hours 6 minutes).

Processing

The audio recordings were first processed by the LENA proprietary software (Xu, Yapanel, & Gray, 2009), creating algorithmic measures such as conversational turn count and adult word count. Each recording was then run through an in-house automated sampler that selected 15- non-overlapping 5-minute segments, randomly distributed across the duration of the recording. Each segment consists of 2 core minutes of annotated time, with 2 minutes of listenable context preceding the annotation clip and 1 minute of additional context following. Because these segments were sampled randomly, across participants roughly 27% of the random 2-minute coding segments contained no speech at all. For questions of *how much does a phenomenon occur*, random sampling schemes can help avoid overestimating speech in the input, but for questions of input *content*, randomly selected samples may be too sparse (Pisani, Gautheron, & Cristia, 2021).

Therefore, we chose to annotate 5 additional (non-overlapping) 2-minute segments specifically for their high density of speech. To select these segments of dense talk, we first conducted an automated analysis of the audio file using the voice type classifier for child-centered daylong recordings (Lavechin, Bousbib, Bredin, Dupoux, & Cristia, 2021) which identified segments likely containing human speech. The entire recording was divided into 2-minute chunks, each ranked highest to lowest by the total duration of the speech segments contained within the chunk. We annotated the 5 highest-ranked segments of each recording. These high-volubility segments allow us to more closely compare our findings to studies classifying the input during structured play sessions, which paint a denser and

differently-proportioned makeup of the language input (Bergelson et al., 2019). In sum, 30 minutes of randomly-sampled input and 10 minutes of high-volubility input (40 minutes total) were annotated per child.

Annotation

Manual annotation of the selected segments was conducted using the ELAN software (Brugman & Russel, 2009). Trained annotators listened through each 2-minute segment plus its surrounding context and coded it using the ACLEW annotation scheme (Soderstrom et al., 2021). For more information about this scheme, see the ACLEW homepage. Speech by people other than the target child was transcribed using an adapted version of the CHAT transcription style (MacWhinney, 2019; Soderstrom et al., 2021). Because the majority of target children in the project are pre-lexical, utterances (e.g. babble) produced by the target child are not yet transcribed. Speech was then further classified by the addressee of each utterance: child, adult, both an adult and a child, pets or other animals, unclear addressee, or a recipient that doesn't fit into another category (e.g., voice control of Siri or Alexa, prayer to a metaphysical entity).

Manual Annotation Training and Reliability. All annotators are tested on the ACLEW scheme prior to beginning corpus annotation, until they reach 95% agreement or better with a “gold standard” coder for segmentation and utterance classification. Training often takes upwards of 20 hours of annotation practice. Following the first pass by annotators, all files were reviewed by a highly-trained “superchecker” to ensure consistency between coders and check for errors. Over a span of three years, 15 trained annotators contributed to this dataset. Ten percent of clips were re-transcribed to assess reliability; further reliability data are provided in corresponding sections below.

Extracting Measures of Language Input

To go from our dimensions of interest (word count, interactiveness, linguistic, conceptual), to quantifiable properties, we used a combination of automated measures

(generated by the proprietary LENA algorithm, Xu et al., 2009) and manual measures (generated from the transcriptions and classifications made by our trained annotators). Altogether, this corpus presently includes approximately 453 hours of audio, 15994 utterances, and 63665 words. LENA measures were calculated over the whole day, and then normalized by recording length. Transcription-based word count and interactiveness analyses were conducted on the random samples only, to capture a more representative estimate. Linguistic and conceptual analyses were conducted on all available annotations in order to maximize the amount of speech over which we could calculate them. These measures are described below and summarized in Table 2.

Automated Word Count. To derive this count, the LENA algorithm segments the recording into clips which are then classified by speaker’s perceived gender (male/female), age (child/adult), and distance (near/far), as well as several non-human speaker categories (e.g., silence, electronic noise). Only segments that are classified as nearby male or female adult speech are then used by the algorithm for its subsequent Adult Word Count (AWC) estimation (Xu et al., 2009). Validation work suggests that this automated count correlates strongly with word counts derived from manual annotations (Cristia, Bulgarelli, & Bergelson, 2020; $r = .71 - .92$, Lehet, Arjmandi, Houston, & Dilley, 2021), and meta-analytic work finds that AWC is associated with children’s language outcomes across developmental contexts (e.g., autism, hearing loss, Wang, Williams, Dilley, & Houston, 2020). Because the recordings varied in length (8 hours 17 minutes to 15 hours 59 minutes), we normalized AWC by dividing by recording length⁴.

Manual Word Count. We also calculated a manual count of speech in the children’s environment. Manual Word Count (MWC) is simply the number of intelligible words in our transcriptions of each child’s recording. Speech that was too far or muffled to be intelligible, as well as speech from the target child and electronic speech (TV, radio, toys) are excluded from this count. Unlike LENA’s AWC, MWC contains speech from other

⁴ To make these measures more comparable, we present both in terms of words per hour.

speakers in the child’s environment (e.g., siblings), not just from adults.

By using automated *and* manual word count, we hope to capture complementary estimates of the amount of speech children are exposed to. While AWC is considered less accurate than manual annotation, it is commonly used due to its ability to readily provide an estimate of the adult speech across the whole day. MWC, because it comes from human annotations, is the gold-standard for accurate speech estimates, but due to feasibility, is only derived from 30 minutes of the recording (sampled in 2-minute clips, at random, as described above).

Interaction.

Conversational Turn Count. One common metric of communicative interaction (e.g., Ganek & Eriks-Brophy, 2018; Magimairaj, Nagaraj, Caballero, Munoz, & White, 2022) is conversational turn count (or CTC), an automated measure generated by LENA (Xu et al., 2009). Like AWC, a recent meta-analysis finds that CTC is associated with children’s language outcomes (Wang et al., 2020). After tagging vocalizations for speaker identity, the LENA algorithm looks for alternations between adult and target child speech in close temporal proximity (within 5 seconds). This can erroneously include non-contingent interactions (e.g., mom talking to dad while the infant babbles to herself nearby), and therefore inflate the count especially for younger ages and in houses with multiple children (Ferjan Ramírez, Hippe, & Kuhl, 2021). Still, this measure correlates moderately well with manually-coded conversational turns ($r_s=0.28-0.75$, Busch, Sangen, Vanpoucke, & van Wieringen, 2018; Ferjan Ramírez et al., 2021; Ganek & Eriks-Brophy, 2018), and because participants in our sample are matched on both age and number of siblings, CTC overestimation should not be biased towards either group.

Proportion of Child-Directed Speech. Our other measure of interaction is the proportion of utterances that are child-directed, derived from the manual annotations. Each proportion was calculated as the number of utterances (produced by someone *other* than the target child) tagged with a child as the addressee, out of the total number of utterances.

Annotator agreement for addressee was 93%, with a kappa of 0.90 [CI: 0.89–0.91].

Linguistic Features.

Type-Token Ratio. As in previous work (e.g., Montag, Jones, & Smith, 2018; Pancsofar & Vernon-Feagans, 2006; Templin, 1957), we calculated the lexical diversity of the input by dividing the number of unique words by the total number of words (i.e., the type-token ratio). Because the type-token ratio changes as a function of the number of words in a sample (Montag et al., 2018; Richards, 1987), we first standardized the size of the sample by cutting the manual annotations in each recording into 100-word bins. We then calculated the type-token ratio within each of these bins by dividing the number of unique words in each bin by the number of total words (~100) and then averaged the type-token ratio across bins for each child⁵. This provided a measure of lexical diversity: per 100 words, how many unique words are children exposed to?

MLU. We also analyzed the syntactic complexity of children’s language input, approximated as mean utterance length in morphemes. Each utterance in a child’s input was tokenized into morphemes using the ‘morphemepiece’ R package (Bratt, Harmon, & Learning, 2022). We then calculated the mean length of utterance (number of morphemes) in each audio recording. We manually checked utterance length in a random subset of 10% of the utterances (n = 2826 utterances), which yielded a intra-class correlation coefficient of 0.94 agreement with the morphemepiece approach (CI: 0.94–0.95, $p < .001$), indicating high consistency.

Conceptual Features. Our analysis of the conceptual features aims to measure whether the extent to which language input centers around the “*here and now*”: things that are currently present or occurring that a child may attend to in real time. We approximate *here-and-nowness* using lexical and morphosyntactic properties of the input.

⁵ Computing TTR over the entire sample instead of averaging over 100-word bins rendered the same pattern of results.

Proportion of temporally displaced verbs. We examined the displacement of events (focusing on the “now” aspect of here-and-now) discussed in children’s linguistic environment, via properties of the verbs in their input. We are attempting to highlight semantic features of the language environment with a morphosyntactic proxy. We do so here by categorizing utterances based on the syntactic and morphological features of verbs, since these contain some time information in their surface forms. We assigned each utterance a temporality value: utterances tagged “displaced” describe events that take place in the past, future, or irrealis space, while utterances tagged “present” describe current, ongoing events. This coding scheme roughly aligns with both the temporal displacement and future hypothetical categories in (Grimminger, Rohlfing, Lüke, Liszkowski, & Ritterfeld, 2020; see also: Hudson, 2002; Lucariello & Nelson, 1987). We do not consider whether any of the noun referents in an utterance are present or attended to by the child, but whether the events concerning them are presently occurring and salient.

To do this, we used the `udpipe` package (Wijffels, 2023) to tag the transcriptions with parts of speech and other lexical features, such as tense, number agreement, or case inflection. To be marked as present, a verb either had to be marked with both present tense and indicative mood, or appear in the gerund form with no marked tense (e.g. ‘you talking to Papa?’). Features that could mark an utterance as displaced included past tense, presence of a modal, presence of ‘if’, or presence of ‘gonna’/‘going to’, ‘have to’, ‘wanna’/‘want to’, or ‘gotta’/‘got to’⁶, since these typically indicate future events, belief states and desires, rather than real-time events. In the case of utterances with multiple verbs, we selected the features from the first verb or auxiliary, as a proxy for hierarchical dominance. Utterances without verbs were excluded. A small number of verb-containing utterances in our corpus were left

⁶ Only the “-to” forms of these verbs are pulled specifically into the “displaced” category, because they specifically select phrasal complements. Sentences like “I want that ball” are treated as having a separate verb than “wanna;” in this case the utterance would be tagged as present tense and put into the “present” category since it is grounded in present objects and events.

“ambiguous” ($n = 1440/8930$), either because they were fragments or because the automated parser failed to tag any of the relevant features. We manually checked verb temporality in a random subset of 10% of the utterances ($n = 825$). Notably, we did not simply verify whether the tagger accurately identified tense and aspect. Rather, human coders holistically tagged the utterance as decontextualized or not, factoring in meaning, context, and syntax, providing a stronger test of reliability against the tagger’s verb-tense based assessment. Human judgments of event temporality aligned with the automated tense tagger 76% of the time ($\text{Kappa} = 0.56$, $\text{CI}:0.56\text{-}0.62$, $p = .050$), indicating substantial agreement, with the majority of discrepancies occurring on words the tagger categorized as ambiguous.

Proportion of highly visual words. In addition to this general measure of decontextualized language, we include one measure that is uniquely decontextualized for blind children: the proportion of words in the input with referents that are highly and exclusively visual. We first filter the input to only content words (excluding, for example: *the*, *at*, *of*). We then categorize the perceptual modalities of words’ referents using the Lancaster Sensorimotor Norms, which are ratings from sighted adults noting the extent to which a word evokes a sensory experience in a given modality (Lynott, Connell, Brysbaert, Brand, & Carney, 2020). Each of the approximately 40,000 words in the Lancaster Sensorimotor Norms gets a score for each of 6 sensory modalities (auditory, haptic, gustatory, interoceptive, olfactory, visual). In this rating system, words with higher ratings in a given modality are more strongly associated with perceptual experience in that modality, and a word’s dominant perceptual modality is the modality which received the highest mean rating. We tweak this categorization in two ways: we categorized content words that received relatively low ratings across all modalities ($<3.5./5$) as predominantly *amodal*, and content words whose ratings were distributed across modalities were categorized as *multimodal*⁷. Using this system, each of the content words in children’s input were

⁷ Words with perceptual exclusivity scores < 0.5 (calculated as a word’s range of ratings across modalities divided by the sum of ratings across modalities, Lynott et al., 2020) were re-categorized as multimodal. The

categorized into their primary perceptual modality; 76% of the words in our corpus had a corresponding word in the Lancaster ratings and could be categorized in this way. For each child, we extracted the proportion of exclusively “visual” words in their home speech sample. Examples of visual words, based on this classification system include: “blueprint”, “see”, “color”, “sky”, “pictures”, “lighting”, “moon”, “glowing”.

Results

Measuring Properties of Language Input

Our study assesses whether language input to blind children is different from the language input to sighted children, along the dimensions of word count, interaction, linguistic, and conceptual properties. We test for group differences using paired t-tests or non-parametric Wilcoxon signed rank tests, when a Shapiro-Wilks test indicates that the variable is not normally distributed (summarized in Table 2). Because this analysis involves multiple tests against the null hypothesis (*that there is no difference in the language input to blind vs. sighted kids*), we use the Benjamini-Hochberg correction (Benjamini & Hochberg, 1995) to control false discovery rate ($Q = .05$) for each set of analyses (word count, interaction, linguistic, conceptual). Because each dimension’s analysis consists of two statistical tests, our Benjamini-Hochberg critical values were $p < 0.025$ for the smaller p value and $p < 0.05$ for the larger p value. The results of these analyses are summarized in Table 2.

Overall Quantity. We first compare the language input to blind and sighted children using two measures of the number of words in their environment: LENA’s automated Adult Word Count and our transcription-derived Manual Word Count. Despite

cut-offs for classifying amodal and multimodal words were chosen based on authors’ intuitions regarding what thresholds seemed to classify the words well into amodal, multimodal, and visual phenomena. That said, results are robust across a range of thresholds, and all data are provided to interested readers should they be interested in considering other values.

Word Count Measures

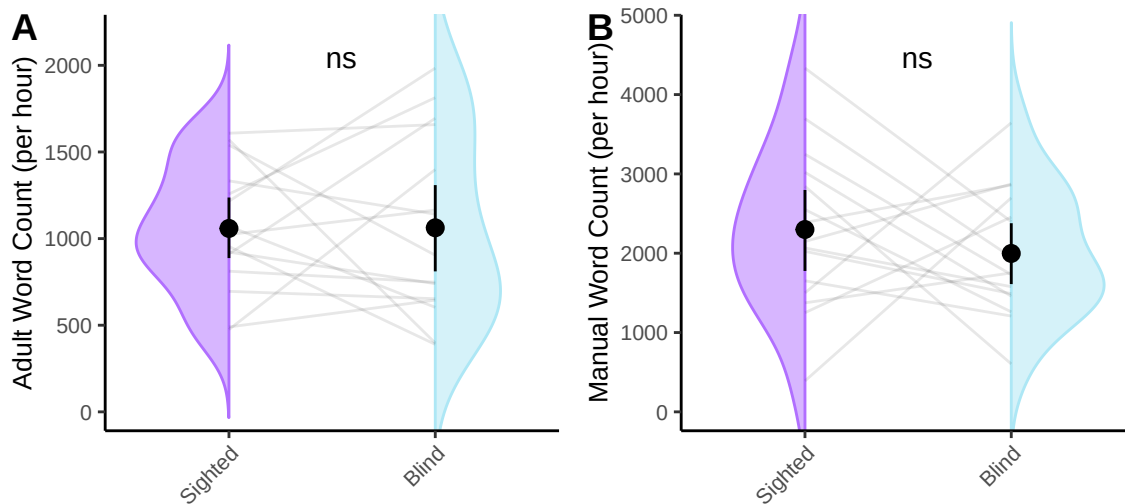


Figure 1. Comparing LENA-generated adult word counts (left) and transcription-based word counts in the input of blind and sighted children. Violin density represents the distribution of word counts for each group. Grey lines connect values from matched participants. Black dot and whiskers show standard error around the mean. Neither measure differed between groups.

wide variability in the number of words children hear (Range from Manual Word Count: 604–3644 words_{blind}, 390–4334 words_{sighted} per hour), along both word count measures, blind and sighted children do not differ (Adult Word Count: $t(14) = -0.02$, $p = .984$; Manual Word Count: $t(14) = 1.06$, $p = .307$); see Figure 1.

Interaction. Our corpus also revealed no significant difference in amount of interaction with the child, measured as the proportion of child-directed speech ($t(14) = 0.24$, $p = .811$) or in conversational turn counts to blind children versus to sighted children ($W = 61$, $p = .978$). Across both groups, child-directed speech constituted approximately 56% of the input, and children were involved in an estimated 34 conversational turns per hour (based on the LENA automated metric); see Figure 2.

Linguistic Features. Similarly, neither linguistic variable differed across groups: blind and sighted children’s input had comparable type-token ratios ($t(14) = -0.96$, $p = .353$)

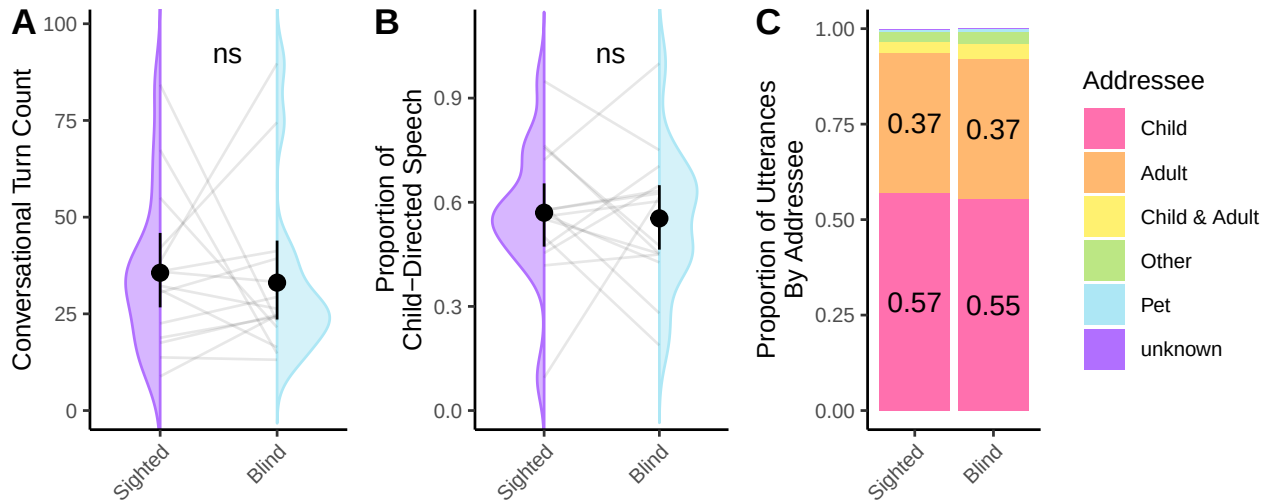
Interaction Measures

Figure 2. Comparing LENA-generated conversational turn counts (left) and proportion of utterances in child-directed speech (center). Violin density represents the distribution of values for each group. Grey lines connect values from matched participants. Black dot and whiskers show standard error around the mean. The full breakdown by addressee is shown in the rightmost panel. Neither conversational turn count nor proportion of child-directed speech differed between groups.

and utterance lengths ($t(14) = -2.02, p = .063$). Children in our samples hear on average 64 unique words per hundred words and 5.20 morphemes per utterance; see Figure 3.

Conceptual Features. Lastly, we compared two measures of the conceptual features of language input: the proportion of temporally displaced verbs and the proportion of highly visual words; see Figure 4. We found that blind children hear a higher proportion of displaced verbs than sighted children ($t(14) = -2.68, p = .018$), which on average equates to 22 more utterances about past, future, or hypothetical events per hour. We found no significant difference across groups in the proportion of highly visual words⁸ ($W = 75, p = .421$), which constitute roughly 10% of the input for both groups.

⁸ And similarly, there were no significant group differences in the proportions of auditory words, tactile words, or non-visual-but-still-perceptual words.

Linguistic Measures

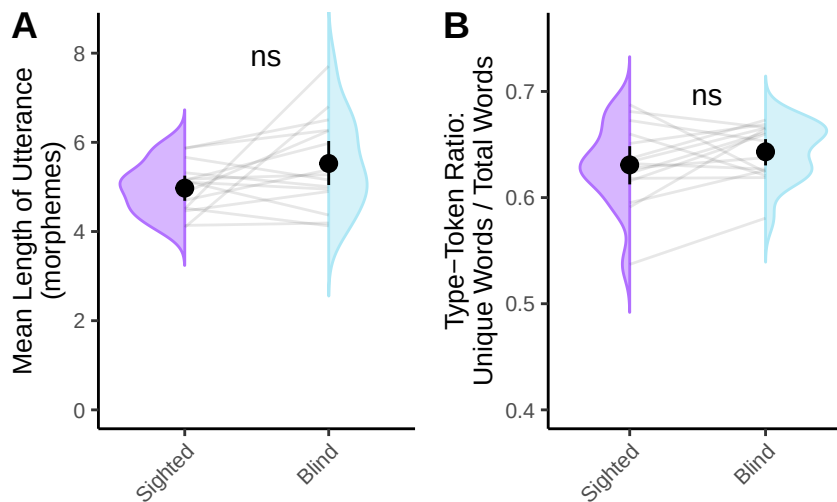


Figure 3. Comparing linguistic features: Mean length of utterance (left) and type-token ratio (right). Violin density represents the distribution of values for each group. Grey lines connect values from matched participants. Black dot and whiskers show standard error around the mean. Utterances in blind children’s input were significantly longer, and type-token ratio was significantly higher. Note that the y-axis on the type-token ratio plot has been truncated.

Evidence of Absence? To explore the extent to which any observed lack of

difference could be interpreted as equivalence – that blind and sighted children’s input did not differ–, we also conducted equivalence tests. Specifically, for variables that did not differ significantly across groups (i.e., adult word count, manual word count, conversational turn count, proportion of child-directed speech, type-token ratio, MLU, and proportion of visual words), we conducted two one-sided equivalence tests (Lakens, 2017) against a small, moderate, and large effect sizes (Cohen’s $d < |0.3|$, $d < |0.5|$, $d < |0.7|$, respectively). Given our relatively small sample, for all but the largest effect sizes tested, results were inconclusive, i.e. it remains possible there are small to moderate differences in the input across the blind and sighted groups. For adult word count, conversational turn count, proportion of child-directed speech, and proportion of highly visual words, we did find a significant equivalence test when the Cohen’s D threshold is set at $|0.7|$.

Conceptual Measures

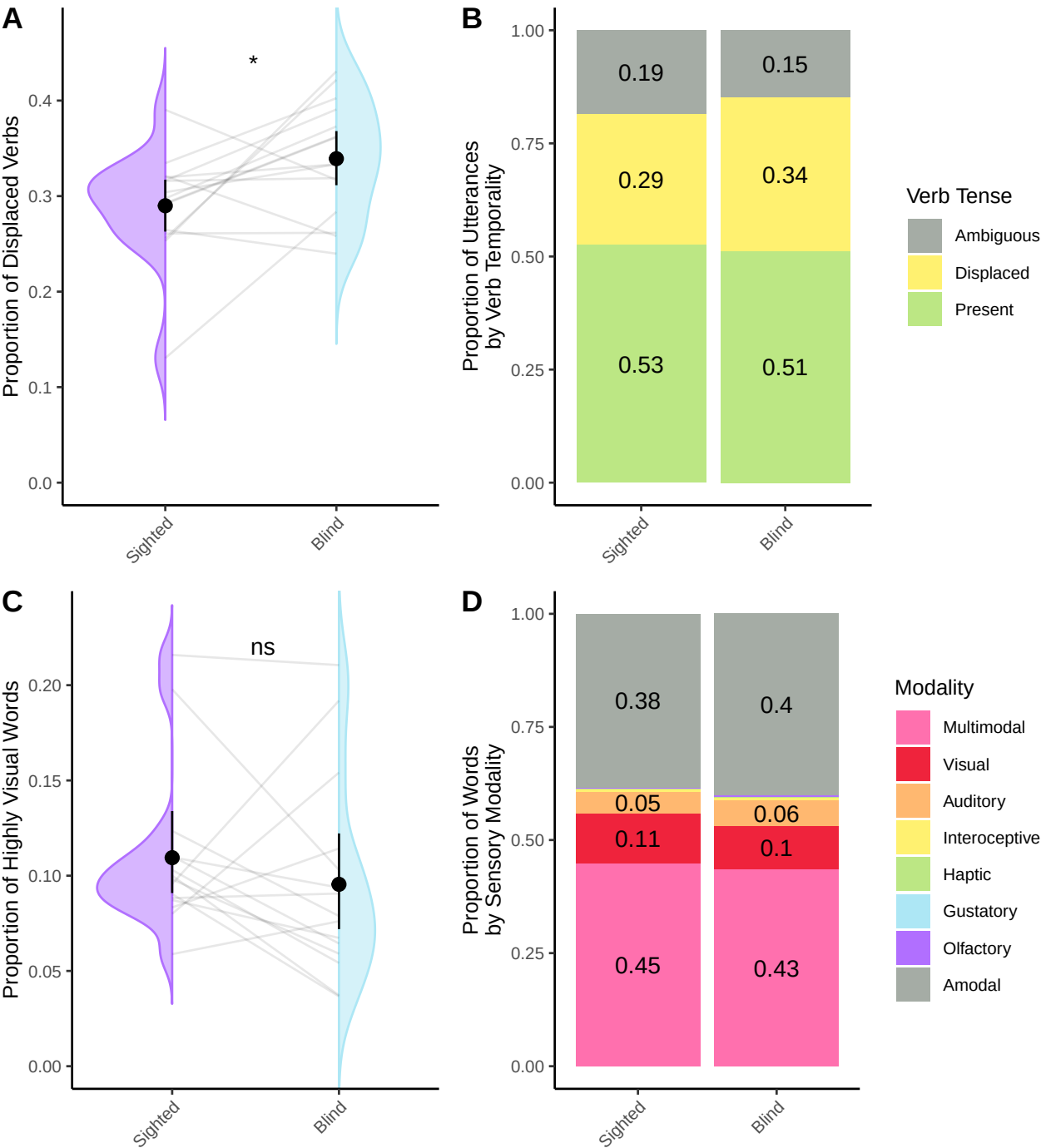


Figure 4. Left col: Comparing proportion of temporally displaced verbs (top) and proportion of highly visual words (bottom). Violin density represents the distribution of values for each group. Grey lines connect values from matched participants. Black dot and whiskers show standard error around the mean. Right col: Full distribution of verb types (top) and sensory modality (bottom) by group, collapsing across participants. Blind children’s input contained significantly more temporally displaced verbs. Notably, the groups did not differ in the proportion of highly visual words.

Age Differences. Lastly, we used a series of linear models (each predicting one of our input variables, based on an interaction between age and group) to explore whether input characteristics may differ for younger vs. older children. We note that these analyses are extremely exploratory, but are offered in the spirit of transparency to comment on the developmental trends in a limited sample. For the number of words in the input, the proportion of child-directed speech, MLU, and the proportion of temporally displaced verbs, we do not find that the input differs across age for either of our groups (all $ps > .05$). We do find that the number of conversational turns increases across age, such that for each month older, children take part in 1.78 more conversational turns per hour ($p = .004$), and this effect does not differ across groups. The proportion of visual words in children's input increases across developmental time for sighted children (by $\sim 0.72\%$ per month, $p < .001$) but not for blind children. An opposite pattern is observed for amodal words: across developmental time, sighted children have fewer amodal words in their input (-0.39% fewer per month, $p < .001$) whereas blind children have marginally more (by $\sim 0.26\%$ per month, $p = .082$). This interaction with age is not observed for any of the other sensory modalities. Tables and figures for these exploratory models are available in Supplementals.

Discussion

In this study, we analyzed the everyday language input to 15 young congenitally-blind children alongside a carefully peer-matched sighted sample using LENA audio recorders. While still relatively modest in absolute terms, this is a larger and more naturalistic sample than has previously been leveraged by prior work with this low-incidence population. We found that along the word count, interaction, and linguistic dimensions, caregivers talked similarly to blind and sighted children, with small but potentially notable differences in conceptual content of the input. We discuss each of these results further below.

Word Count

Across two measures of input word count, one estimated from the full sixteen hour recording (Adult Word Count) and one precisely measured from a 30-minute samples from the day (Manual Word Count), blind and sighted children were exposed to similar amounts of speech in the home. Word count was highly variable *within* groups, but we found no evidence for *between* group differences, though it remains a possibility that there are smaller effects that we were unable to detect. This lack of difference runs counter to two folk accounts of language input to blind children: 1) that sighted parents of blind children might talk *less* because they don't share visual common ground with their children; 2) that parents of blind children might talk *more* to compensate for their children's lack of visual input. Instead, we find a similar amount of speech across groups.

Interaction

We quantified interaction in two ways: through the LENA-estimated conversational turn count and through the proportion of child-directed speech in our manual annotations. Again, we found no differences across groups in the amount of parent-child interaction. This finding contrasts with previous research; other studies tend to report *less* interaction in dyads where the child is blind (Nagayoshi et al., 2017; Rogers & Puchalski, 1984; as measured by responsiveness, Tröster & Brambring, 1992; initiations of interactions, Andersen et al., 1993, 1993; Dote-Kwan, 1995; Kekelis & Andersen, 1984; Moore & McConachie, 1994; Tröster & Brambring, 1992; caregiver dominance of the conversation, Kekelis & Andersen, 1984; or "weak and inconsistent" responses to blind infants' vocalizations, Charity Rowland, 1984). Our use of daylong audio recordings might explain this apparent discrepancy in results. For one thing, many prior studies (e.g., Kekelis & Andersen, 1984; Moore & McConachie, 1994; Pérez-Pereira & Conti-Ramsden, 2001; Preisler, 1991) involve videorecordings in the child's home, with the researcher present. Like other young children, blind children distinguish between familiar individuals and strangers, and react with

505 trepidation to the presence of a stranger; for blind children, this reaction may involve
 506 “quieting”, wherein children cease speaking or vocalizing when they hear a new voice in the
 507 home (Fraiberg, 1975; McRae, 2002). By having a researcher present during the recordings⁹,
 508 prior research may have artificially suppressed blind children’s initiation of interactions.
 509 Even naturalistic, observer-free videorecordings appear to inflate aspects of parental speech,
 510 relative to daylong audio recordings (Bergelson et al., 2019).

511 Additionally, a common focus in earlier interaction literature is to measure visual cues
 512 of interaction, such as shared gaze or attentiveness to facial expressions (Baird, Mayfield, &
 513 Baker, 1997; Nagayoshi et al., 2017; Preisler, 1991; Rogers & Puchalski, 1984). For example,
 514 Nagayoshi et al. (2017) write: “Infants with visual impairment were characterized by high
 515 likelihood of developmental delays and problematic behaviors; they tended not to turn their
 516 face or eyes toward their mothers.” We can’t help but wonder: are visual markers of social
 517 interaction the right yardstick to measure blind children against? In line with MacLeod and
 518 Demers (2023), perhaps the field should move away from sighted indicators of interaction
 519 “quality”, and instead situate blind children’s interactions within their own developmental
 520 niche, one that may be better captured with auditory- or tactile-focused measures. While
 521 daylong audio recordings excel at capturing extended, naturalistic spoken language use, they
 522 miss non-verbal information, like proximity, touch, or physical properties of the referent. In
 523 contrast, video recordings could provide rich information about these multimodal features.
 524 Future work should consider integrating these approaches to provide a more comprehensive
 525 view of blind children’s interactions.

526 **Linguistic Features**

527 Along the linguistic dimension, we measured type-token ratio and mean length of
 528 utterance. Parents of children with disabilities (including parents of blind children, e.g.,

⁹ Fraiberg (1975) writes “these fear and avoidance behaviors appear even though the observer, a twice-monthly visitor, is not, strictly speaking, a stranger.” (pg. 323).

Chernyak, n.d.; FamilyConnect, n.d.) are often advised to use shorter, simpler sentences with their children; correspondingly, previous work finds that parents of children with disabilities tend to find that parents *do* use shorter, simpler utterances (e.g., Down syndrome, Lorang et al., 2020; hearing loss, Dirks et al., 2020). While language input patterns among these populations may not necessarily generalize to blind children, the societal infantilization of disabled people broadly, including blind individuals (Bulk et al., 2020; Hernandez Padilla & Arias Valencia, 2024), might lead to differences in how caregivers structure their input. We therefore hypothesized that caregivers might provide shorter utterances and less lexically diverse input to blind children compared to their sighted peers. Instead, we found that blind children heard indistinguishable input by these metrics, with, if anything, a (marginally significant) trend towards *longer* sentences in their input (relative to sighted matches, roughly half a word longer on average). Returning to the potential impact of input properties on children, evidence suggests that (contrary to the advice often given to parents), longer, more complex utterances are associated with better child language outcomes in both typically-developing children (Hoff & Naigles, 2002) and children with cognitive differences (Sandbank & Yoder, 2016). And similarly, higher lexical diversity is associated with larger vocabulary (Anderson et al., 2021; Hsu et al., 2017; Huttenlocher et al., 2010; Rowe, 2012; Weizman & Snow, 2001). Regardless, the present analysis did not reveal robust statistical evidence that, at least on the group level, caregivers systematically provide utterances with different length or lexical diversity as a function of whether their child could see.

Conceptual Features

Although there are many potential ways to measure the conceptual features of language, we chose to capture *here-and-now*-ness by measuring the proportion of temporally displaced verbs (i.e., targeting non-present events) and the proportion of highly visual words. We found that blind children heard roughly 5% more temporally displaced verbs than sighted peers. This measure is imperfect: in using tense as a proxy for conceptual features, it

fails to adjudicate, for example, the decontextualized nature of a “make-believe” utterance in the present tense, or the presence of a past-tense utterance describing an event that happened seconds before¹⁰. Nonetheless, we believe this captures similar or higher amounts of signal as much more costly manually-annotated measures. Moreover, though blind and sighted participants were exposed to a similar proportion of highly visual words, the referents of these words are by definition inaccessible to the blind participants. Taken together, our conceptual results suggest that blind children’s input could be *less* focused on the *here-and-now*.

The extent to which blind children’s language input is centered on the *here-and-now* has been contested in the literature (Andersen et al., 1993; J. Campbell, 2003; Kekelis & Andersen, 1984; Moore & McConachie, 1994; Urwin, 1984). This aspect of language input is of particular interest because early reports suggest that blind children’s own use of decontextualized language develops later than sighted children’s (Bigelow, 1990; Urwin, 1984). Could such a difference be attributable to an absence of decontextualized language in the input? Our results suggest this is unlikely: we find that blind children’s input contains *more* decontextualized language (as indicated by verb temporality) rather than less. Speculatively, this may be because visually-oriented, sighted caregivers find a perceptual common ground for discussion, instead of replacing visually-grounded conversation with sensory modalities that the child can access. For example, while riding on a train, parents of sighted children may discuss the changing scenery outside the window, which is present, perceptually accessed by both parent and child, and salient as a topic of conversation. Present, perceptually available features of the environment for the blind child, such as the

¹⁰ One concern about this metric is its treatment of multi-clause utterances. For example, in “I went to the grocery store and now I’m watching TV”, “went” is not syntactically higher than “watching” but our classification system would rely on the tense of “went” alone. In practice, only 1.7% of utterances in our dataset contain verbs both before and after a conjunction, while 11.05% contain syntactic subordination, where the tense of the highest verb is most appropriate to assess.

rumble of the train and velvety feel of the seats, may be less salient to the sighted parent as a topic of discussion, which may lead the caregiver to choose to talk about events that happened earlier in the day or their plans upon arriving home. Past and future events are experienced via mental representation rather than perceptually for caregiver and child alike. This is a potential avenue for broadening the concept of joint attention as a fundamental feature of conversation and language acquisition beyond shared visual reference.

Our findings indicate that sighted caregivers use a comparable amount of ‘highly visual’ words when speaking to their blind children and their sighted peers, as measured using sensorimotor norms derived from sighted adults (Lynott et al., 2020). While these norms offer a valuable framework for analyzing input from sighted caregivers, it is important to consider the semantic implications for blind children themselves. Kerr and Johnson (1991) reported that blind adults rated traditionally visual words, like ‘sky,’ as evoking more varied and multimodal mental imagery, including tactile and spatial experiences. Future work developing sensory norms specifically tailored to blind individuals would provide valuable insights into these children’s perceptual-semantic mappings. In the meantime, our findings suggest that while caregivers do not reduce their use of visual words when interacting with blind children, these words could potentially take on unique semantic dimensions within the linguistic and sensory environments of blind learners.

Without further information about the social and perceptual context, it is difficult to determine the motivation of the differences we find in conceptual features we find. As more dense annotation becomes available, we look forward to further work exploring the social and environmental contexts of conceptual information as it unfolds across discourse.

It is worth underscoring again how much variability there is *within* groups and how much consistency there is *between* groups. One could imagine a world in which the language environments of blind and sighted children are radically different from each other. Our data do not support that hypothesis. Rather, we find similarity in word count, interaction, and

linguistic properties, alongside modest differences in conceptual properties. That is, in line with recent work highlighting immense *within*-group variability across many different socio-cultural and linguistic contexts (Bergelson et al., 2023), our blind and sighted groups here have large within-group variability but very few between-group differences. Despite strikingly different visual experiences, young blind and sighted learners have at best modest differences in their speech environments.

Connecting to Language Outcomes

Our results uncover no systematic group differences in word count, amount of language interaction, or linguistic complexity parents provide to blind vs. sighted children, at least as measured here. When we do see differences, language input to blind children looks more conceptually complex or perceptually unavailable. In other populations, complexity of this sort is linked with *more* sophisticated child language outcomes (Demir et al., 2015; Rowe, 2012; Uccelli et al., 2019), so it is not the case that blind children’s language input is “impoverished” in this sense.

In our modestly-sized, predominantly pre-lexical sample, linking language input to children’s language outcomes directly is not yet feasible, but prior literature allows us to speculate on two possibilities. First, if input effects pattern similarly for blind and sighted children, we would expect blind and sighted children alike to benefit from more input (Anderson et al., 2021; Gilkerson et al., 2018; Huttenlocher et al., 1991; Rowe, 2008), more interactive input (Donnellan et al., 2020; Goldstein & Schwade, 2008; Hirsh-Pasek et al., 2015; Romeo et al., 2018; Rowe, 2008; Shneidman et al., 2013; Weisleder & Fernald, 2013), more linguistically complex input (Anderson et al., 2021; De Villiers, 1985; Hadley et al., 2017; Hoff, 2003; Hsu et al., 2017; Huttenlocher et al., 2002, 2010; Naigles & Hoff-Ginsberg, 1998; Rowe, 2012; Weizman & Snow, 2001), and more conceptually complex input (Demir et al., 2015; Rowe, 2012; Uccelli et al., 2019).

At the same time, however, recent results show that blind children have a roughly half-year delay in their productive vocabulary, relative to sighted peers (E. E. Campbell et al., 2024). If properties of the language input play a role in this delay, this raises the second possibility: that language input affects acquisition *differently* for blind children than it does for sighted children. Under this possibility, blind children would benefit from *less* complex language input, and the equivalencies in word count, linguistic complexity and interactivity alongside the increased conceptual complexity we find here would, in theory, contribute to early vocabulary delays.

To show our cards, we are inclined towards option one: that blind children benefit from language input in the same ways as sighted peers (Landau & Gleitman, 1985), and that this additionally extends to the benefits of receiving more conceptually complex language input. Language regularly supports learning in the absence of direct sensory perception (e.g., reading a book about mythical creatures). Given the language skills of blind adults (Loiotile et al., 2020; Röder et al., 2003; Röder et al., 2000), it is undeniable that language is a rich source of meaning for blind individuals as well (E. E. Campbell & Bergelson, 2022; Lewis, Zettersten, & Lupyan, 2019; van Paridon, Liu, & Lupyan, 2021). Testing each of these predictions—as well as whether links between language input and language outcomes change across developmental time—awaits further research.

In either case, if properties of language input do influence blind children’s language outcomes, attempting to train parents to talk differently may be unfruitful. While some interventions where parents are trained to talk differently to their children show promise (Huber, Ferjan Ramírez, Corrigan, & Kuhl, 2023; Roberts, Curtis, Sone, & Hampton, 2019), such interventions often fail to change parental speech patterns on more extended timescales (e.g., McGillion, Pine, Herbert, & Matthews, 2017; Suskind et al., 2016).

Conclusion

In summary, our study compared language input in homes of 15 blind and 15 sighted infants. We found that both groups received language input with similar quantities of speech, interactivity, and linguistic complexity. Additionally, blind children were exposed to input that had somewhat more conceptual complexity, in terms of discussion beyond the here-and-now and words for less perceptually-available (visual) referents. This suggests that young blind children are being exposed to a rich linguistic environment that differs only modestly from the language input of sighted children. Our study does not imply that parents should change their communication styles, but rather highlights the language experiences of blind children. Future research linking input links to language development and cognitive abilities of blind and sighted children alike would be a fruitful and welcome next step.

Ethics

This study received approval from the Duke University Institutional Review Board. All families consented to take part in this research.

Data, Code and Materials Availability

LENA data, transcripts, and code for all analyses presented in this article are available on OSF (<https://osf.io/dcnq6/>).

Authorship

Erin Campbell: Conceptualization, Validation, Investigation, Analysis, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project Administration; **Lillianna Righter:** Validation, Analysis, Data Curation, Writing - Original Draft, Writing - Review & Editing, Project Administration; **Eugenia Lukin:** Validation, Writing - Review & Editing; **Elika Bergelson:** Conceptualization, Resources, Writing - Review & Editing, Visualization, Supervision, Funding Acquisition

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Table 2

Summary of language input variables: how the measure is calculated; what portion of the recording the measure was calculated over; whether a parametric or non-parametric test was used; the mean, median, and range for blind and sighted children, and the raw (uncorrected) p-value of the test comparing groups. Only prop. displaced reached significance at our corrected $p < .025$ threshold for significance.

Variable	Description	Portion of Recording	Test	Blind Mean, Median, Range	Sighted Mean, Median, Range	p value
Adult Word Count	Estimated number of words in recording categorized as nearby adult speech by LENA algorithm	Whole day	t-test	2124, 1808, 779-3968 words/hour	2117, 2047, 951-3216 words/hour	.984
Manual Word Count	Number of word tokens from speakers other than target child	Random	t-test	3994, 3504, 1208-7288 words/hour	4598, 4296, 780-8668 words/hour	.307
Conversational Turn Count	Count of temporally close switches between adult and target-child vocalizations, divided by recording length	Whole day	Wilcoxon test	66, 49, 26-180 turns/hour	71, 65, 18-169 turns/hour	.811
Prop. Child-Directed Speech	Number of utterances tagged with child addressee out of total number of utterances, from speakers other than target child	Random	t-test	0.55, 0.6, 0.19-1	0.57, 0.57, 0.09-0.95	.978
Type-Token Ratio	Average of the type-token ratios (number of unique words divided by number of total words) for each of the 100-word bins in their sample	Random + High Volume	t-test	0.64, 0.65, 0.58-0.67 unique words/ 100 words	0.63, 0.63, 0.54-0.69 unique words/ 100 words	.353
Mean Length of Utterance	Average number of morphemes per utterance	Random + High Volume	t-test	5.53, 5.28, 4.13-7.71 morphemes	4.97, 5.11, 4.09-5.87 morphemes	.063
Prop. Displaced	Proportion of verbs that refer to past, future, or hypothetical events	Random + High Volume	t-test	0.34, 0.33, 0.24-0.43	0.29, 0.3, 0.13-0.39	.018*
Prop. Visual	Proportion of words in the input with high visual association ratings and low ratings for other perceptual modalities	Random + High Volume	Wilcoxon test	0.1, 0.08, 0.04-0.21	0.11, 0.1, 0.06-0.22	.421